

Rithani Saravanakumar

rithanisk@gmail.com | [linkedin.com/in/rithanisk](https://www.linkedin.com/in/rithanisk) | github.com/rithanisk | <https://rithanisk.vercel.app>

EDUCATION

National University of Singapore

Expected Graduation: May 2027

Bachelor of Computing (Honours) in Computer Science

Relevant Coursework: Data Structures/Algorithms, Operating Systems, Software Engineering, AI/Machine Learning

SKILLS

Languages: Golang, Python, Java, C/C++, R, JavaScript, Typescript, SQL

Frameworks/Libraries: FastAPI, Next.js, TensorFlow, Keras, LangChain, Agno

Technologies/Tools: Git, IntelliJ, Postgres, MySQL, MongoDB, Docker, Pinecone, Hugging Face, AWS, Azure

EXPERIENCE

AI/Software Engineer

August 2025 - July 2026

NiaHealth

- Reduced medical report parsing latency by **70%** (10→3 min) and saved the COO **6–8 hrs/week** of manual review by architecting an asynchronous, background-worker GPT pipeline for external document ingestion.
- Eliminated **8 critical data errors** before patient-facing delivery, protecting a team of 10 clinicians from potential legal liability, by engineering AI report quality checks using Agno into a clinician-facing internal reporting tool.
- Improved food photo calorie extraction accuracy by **27%** over baseline GPT pipeline by integrating FNDDS nutritional database search into a multiagent AI pipeline and validating through systematic model evaluations.
- Achieved **78% female user engagement** on a net-new cycle tracker by owning the full product lifecycle from requirements gathering with clinical stakeholders through UI mockups, technical spec, and iterative development.

Software Engineer Intern

May 2024 - August 2024

Buymed

- Improved page load time by **25%** and boosted user retention by **10%** across high-traffic web applications by shipping 10+ production features and resolving 15+ critical performance bugs.
- Reduced competitor pricing data collection effort, measured across **6 external pharmaceutical sources**, by building a concurrent web scraping and pricing aggregation pipeline with simultaneous multi-source extraction.
- Decreased customer information retrieval time from **minutes to seconds** by scraping and embedding the company website into a vector database and deploying a user-facing RAG chatbot with hallucination guardrails.

PROJECTS

Google Mediapipe Sign Language Recognition

- Achieved **88.7% classification accuracy** across all 28 single-hand Bharatanatyam mudras by training a custom Google MediaPipe gesture recognition model on 21-point hand landmark data augmented on 1,400+ training samples.
- Enabled **20+ young Bharatanatyam students** to independently practice and self-correct mudra technique in real time by deploying an end-to-end Flask web application with live camera inference.

ML Pipeline

- Achieved 80% AUC-ROC on student dropout prediction by designing and implementing an end-to-end ML pipeline in Python using Pandas, Scikit-learn, and LightGBM, engineering 67 features from merged datasets

Sepsis Mortality Prediction Model

- Identified significant correlations in 40,000+ MIMIC-III patient records and developed an early sepsis mortality prediction model by analyzing large-scale time-series data and building an LSTM neural network with Keras.

NVIDIA Build It Yourself Agentic AI

- Completed NVIDIA's **Build It Yourself Agentic AI** virtual training series, building autonomous goal-driven agents using the NemoClaw™ and OpenShell™ stack across projects including an academic planner, system-aware research assistant, and collaborative multi-agent team.